The Effect of Information Entropy on Recognition Memory for Words

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Introduction: Lexical prediction plays a central role in language comprehension, but its effects on subsequent memory for words remain unclear. Previous studies have shown that predictable words can lead to false memories (Hubbard et al., 2019; Haeuser & Kray, 2022) and that unexpected words often receive deeper semantic processing, which may strengthen encoding (Rommers & Federmeier, 2018). However, little is known about how the distribution of predicted alternatives contributes to subsequent memory. For instance, in "In polluted cities, you can barely breathe the __," readers may predict a few semantically related completions such as *air* or *smog*. In contrast, "They suspected there was something wrong with the __" may elicit many unrelated continuations such as *air*, *software*, *car*, or *water*. These differences reflect variation in predictive uncertainty that is not captured by traditional measures such as cloze probability. Information-theoretic accounts quantify this uncertainty using information (Shannon) entropy (Hale, 2003; 2016). Recent work has demonstrated that entropy includes two distinct components: *Semantic entropy* reflects semantic overlap between the target word (e.g., *air*) and the rest of the responses, and has been shown to facilitate online processing. Lexical entropy reflects competition among lexical candidates, which has been shown to inhibit online lexical processing (Karimi et al., 2025). However, the potential effects of semantic and lexical entropy on subsequent memory remain unexplored. The present study investigates whether and how semantic and lexical entropy differentially affect subsequent recognition memory for words.

Method: 60 native English speakers participated in the study. They first completed a cloze task in which they generated up to 8 completions for 216 sentence fragments and assigned probability estimates to each word (e.g., *air*→ 90%, *smog*→ 40% etc.). These responses and their associated (and normalized) probability values were used to calculate information (Shannon) entropy values per trial (i.e., for each participant, for each item). Fourteen to twenty-one days later, the same participants read the full sentences while their eye movements were recorded. Table 1 shows a sample item across the cloze and reading sessions. Immediately the reading session, they completed a surprise recognition test (Figure 1) containing 216 sentence-final target words and 120 lures. We computed the sum of cosine semantic similarity values between the target words (e.g., *air*), and the rest of the provided responses for that trial (e.g., *smog*) from the cloze task. Then using multiple regression, we predicted recognition accuracy as a function of information (Shannon) entropy and semantic similarity. Thus, any effect of information entropy while semantic overlap (semantic entropy) was controlled should reflect competition among lexical candidates (i.e., lexical entropy). Recognition accuracy was analyzed using logistic mixed-effects models with trial-level information entropy and trial-level semantic similarity. The model also included total reading times on target words (e.g., *air*) during reading as a covariate. Random intercepts and slopes (for entropy and semantic similarity) were included in the model.

Results: As shown in Figure 2 and Table 2, semantic entropy reliably enhanced recognition accuracy ($\beta = 0.20$, $z = 6.19$, $p < .001$), whereas lexical entropy reduced accuracy ($\beta = -0.09$, $z = -3.04$, $p = .002$). Total reading times on the target words did not reliably predict memory performance ($\beta = 0.04$, $z = 1.59$, $p = .11$).

General Discussion: The findings suggest that information entropy influences subsequent memory depending on the semantic and lexical structure of the prediction space. Semantic entropy (overlap) appears to support retrieval by activating features (cues) that overlap with the encoded target word (e.g., *air*). Lexical entropy may impair retrieval by increasing competition among retrieval candidates. The absence of a reliable effect of reading time indicates that memory performance does not depend on processing (encoding) time alone. By separating semantic and lexical components of entropy, this study shows that information entropy shapes not only online comprehension but also the processes through which linguistic information is encoded and later retrieved.

Table 1. *Example Sentence Fragments and Full Sentences Presented in the Cloze Task and Reading Task Under Low, Medium, and High Constraint Conditions*

Constraint Conditions /Task	Cloze session	Reading session
a) Low Constraint	They suspected there was something wrong with the ___.	They suspected there was something wrong with the air outside the building at noon.
b) Medium Constraint	The factory built expensive machines to filter the ___.	The factory built expensive machines to filter the air outside the building at noon.
c) High Constraint	In polluted cities, you can barely breathe the ___.	In polluted cities, you can barely breathe the air outside the building at noon.

Figure 1. *Sample of Surprise Recognition Memory Test*

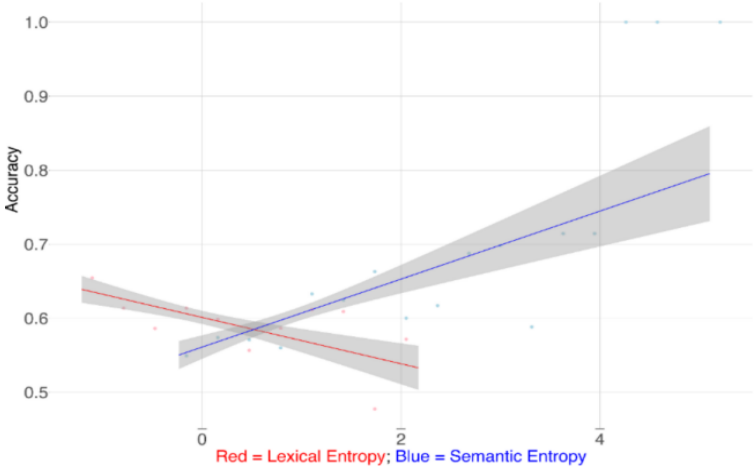

Figure 2. *Effects of semantic entropy and lexical entropy on recognition memory accuracy (both variables were centered). Higher semantic entropy, reflecting richer activation of target-related semantic information, is associated with better recognition performance. In contrast, higher lexical entropy, reflecting stronger competition among predicted candidates, is associated with reduced recognition performance. Shaded regions represent 95% confidence intervals.*

Table 2. *Mixed-Effects Logistic Regression Predicting Recognition Memory Accuracy from Semantic Entropy, Lexical Entropy, and Reading Time*

Predictor	*Estimate*	*SE*	*z*	*p*
Intercept	0.55	0.13	4.22	< .001
Semantic entropy	0.20	0.03	6.19	< .001
Lexical entropy	-0.09	0.03	-3.04	.002
Reading time	0.04	0.03	1.59	.112